Environmental
Resources
Management

3352 128th Avenue
Holland, MI 49424-9263
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http://www.erm.com

ERM

13 February 2017
Reference: 0283400.0106

Mr. Derek Goodrich
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Re: Whole Effluent Toxicity Test Results

Dear Derek:

Enclosed please find the final results of the following Chronic Toxicity Tests performed on samples of the City of Cadillac WWTP Outfall 001A effluent.

- 2 February 2017, Chronic *Ceriodaphnia dubia* Toxicity Test
- 2 February 2017, Chronic *Pimephales promelas* Toxicity Test

If you have any questions concerning this report or if I can be of any further assistance to you, please feel free to contact me at (616) 738-7308 or via e-mail at bruce.rabe@erm.com.

Sincerely,

Bruce A. Rabe

Bruce A. Rabe
Director, Aquatic Toxicology Laboratory

BAR:km
Enclosures: Whole Effluent Toxicity Test Report
cc: Doug Langworthy
File

FINAL REPORT

Chronic Toxicity Test
Freshwater Invertebrate,
Ceriodaphnia dubia
EPA Test Method 1002.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Sample: Outfall 001A

Testing Period: 2 – 8 February 2017

Laboratory I.D. Number: 020217-2

PA DEP ID No. 68-04227
NJ DEP ID No. MI013

Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424

Test Overview

Permittee:	City of Cadillac WWTP
Location:	1121 Plett Rd. Cadillac, MI 49601
Contact:	Mr. Derek Goodrich
Telephone #:	231.878.4437
NPDES Permit #:	MI0020257
Permit Requirements:	Acute Toxicity = Report Only Chronic Toxicity Limit = 1.1 TUc
Test Sample:	Outfall 001A
Receiving Water:	Clam River
Testing Date:	2 - 8 February 2017
Sample Date(s):	1 February 2017 3 February 2017 6 February 2017
Test/Method:	Daphnid, *Ceriodaphnia dubia*, Survival and Reproduction Test EPA 821-R-02-013 Method 1002.0.
QC Objectives:	Test data met all test acceptability criteria, except where noted below.
Data Qualifiers:	None.

DATA SUMMARY

Effluent Concentrations (%)	Survival (%)	Reproduction (Average Young/Female)
Control	100	26.6
13	100	27.6
25	100	32.7
50	100	30.4
76	100	27.2
100	100	27.0

TEST RESULTS

48-Hour LC_{50}	>100%
NOEC	100%
LOEC	>100%
IC_{25}	>100%
MSDp	14.6%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	0

TEST CONCLUSION

In accordance with the NPDES permit requirement for the City of Cadillac WWTP, this toxicity test did not exhibit acute toxicity or exceed the chronic toxicity limit.

Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0343408.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
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ERM

ERM Testing Method

Ceriodaphnia dubia – Survival and Reproduction Toxicity Test

Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (µm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius (°C) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Ceriodaphnia dubia used to initiate this test were obtained from individual, in-house cultures and were less than 24-hours old, and had an age range of 0 to 8 hours at test initiation. Test organisms used to initiate this test were released from adults which met acceptable performance criteria (i.e., ≥15 young/surviving female within 3 broods and obtained from a brood of at least 8 young) and were maintained in reconstituted moderately hard water prior to test initiation.

The *Ceriodaphnia dubia* test was conducted using 30-milliliter (mL) disposable polystyrene containers containing 15 mL of control water or test solution. One *Ceriodaphnia dubia* was added to each test chamber with ten replicate chambers per treatment. Each *Ceriodaphnia dubia* test chamber was fed a 0.2-mL suspension consisting of yeast-Cerophyll-trout chow (YCT) and green algae (*Raphidocelis subcapitata*) mixture daily.

The test solutions were renewed daily during the exposure by transferring the adult daphnid, by way of a wide bore pipette, into fresh control water or test solution.

Percent survival of exposed *Ceriodaphnia dubia* was determined by inspecting for adult mortality daily. Mortality was defined as no body or appendage movement after gentle prodding. Production of young was also determined by daily inspections and enumeration. When 60 percent of the surviving females in the control treatment produced three broods, mean reproduction was determined by calculating the average number of live young produced per female for each treatment.

The test was conducted at a temperature of 25 ± 1°C under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Ceriodaphnia dubia* survival and reproduction. An NOEC is defined as the highest effluent concentration that does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration that does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration. Significant differences in *Ceriodaphnia dubia* survival were determined using the Fisher's Exact Test.

Prior to the determination of any significant differences in *Ceriodaphnia dubia* reproduction, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or reproduction) and converted to a chronic toxic unit (TUc) for reporting purposes by 100/ChV, unless the LOEC was greater than 100% in which case the TUc was reported as 0.

To evaluate acute toxicity, a 48-hour LC_{50} and corresponding 95 percent confidence interval was also calculated, where possible. The LC_{50} value estimate was determined by using one of the following statistical methods: graphical, Spearman-Karber, Trimmed Spearman-Karber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 48-hour LC_{50} value was converted to an acute toxic unit (TUa) by $100/LC_{50}$.

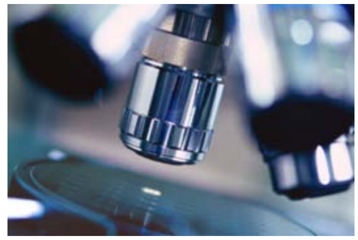
For 48-hour LC_{50} values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 percent to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 percent to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.8.7.15 software program.

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms and the precision of the testing procedure. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC_{25}) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC_{25} of a particular reference toxicant test does not fall within the expected range of ± two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM’S STANDARD TEST METHOD

Test was performed in accordance with ERM’s standard test method (see page 3).

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

Environmental Resources Management

Ceriodaphnia dubia - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation

Permittee/Client:	City of Cadillac WWTP		Control/Dilution Water:	RMHW
Effluent/Location:	Outfall 001A		Organism Batch #:	27-17
Lab I.D.#:	020217-2		Organism Age:	19-21 hrs
Beginning Date:	02/02/17	Time: 1100	QC Review:	KM
Ending Date:	02/08/17	Time: 1130	QC Review Date:	02/10/17

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	020217-2	020417-1	020717-5	06-17	—	—
Temperature	° C	|	|	|	--	--	--
pH	S.U.	6.8	7.1	6.5	7.8	—	—
Conductivity	umhos/cm	1180	1364	1038	317	—	—
Alkalinity	mg / L $CaCO_3$	120	140	100	58	—	—
Hardness	mg / L $CaCO_3$	220	240	460	86	—	—
Total Ammonia	mg / L NH_3	~~70-60~~	2.5 KM 0310 1.3	<0.25	--	--	--
Total Residual Chlorine	mg / L Cl_2	<0.01	<0.01	<0.01	--	--	--
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	CM	SR	KM	CM	—	—

Test Solution Preparation: Test Solution Prepared For Both Species.

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)		Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	1200		0	CM	1	06-17
13%	156	1044		1	CM	1	06-17
25%	300	900		2	SR	2	06-17
50%	600	600		3	SR	2	06-17
76%	912	288		4	CM	~~2~~ 3	06-17 CM 02/6
100%	1200	0		5	RA	3	06-17
				6	CM	3	06-17
				7			

Environmental Resources Management

Ceriodaphnia dubia - Chronic Toxicity Test
Water Quality Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 020217-2

Dissolved Oxygen (mg/L)

Day														
Meter #	3	3	3	3	3	3	3	3	3	5	5	5	5	
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.5	8.0	8.5	8.0	8.2	7.5	8.1	8.0	8.5	7.9	8.4	8.4	8.3	—
13%	8.5	8.1	8.5	8.2	9.3	7.6	8.2	8.0	8.5	7.9	8.4	8.1	8.3	—
25%	8.5	8.1	8.5	8.1	8.3	7.7	8.3	8.0	8.6	7.9	8.2	8.0	8.3	—
50%	8.5	8.1	8.5	8.0	8.4	7.7	8.3	8.0	8.6	7.9	8.3	8.1	8.3	—
76%	8.5	8.1	8.5	7.9	8.5	7.7	8.3	8.0	8.6	7.9	8.4	8.1	8.2	—
100%	8.5	8.0	8.5	8.0	8.6	7.7	8.4	8.0	8.5	7.9	8.4	8.0	8.2	—
														—

pH (S.U.)

Day														
Meter #	9	8	8	8	8	8	8	8	8	9	9	9	9	
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.8	7.7	7.8	7.7	7.6	7.7	7.6	7.6	7.8	7.8	7.9	7.5	7.8	—
13%	--	7.8	--	7.9	--	7.8	--	7.9	--	7.9	--	7.5	--	—
25%	--	7.9	--	8.0	--	7.9	--	8.0	--	8.0	--	7.6	--	—
50%	--	8.0	--	8.0	--	8.0	--	8.0	--	8.1	--	7.6	--	—
76%	--	8.1	--	8.1	--	8.1	--	8.1	--	8.3	--	7.7	--	—
100%	7.2	8.1	7.2	8.2	7.3	8.2	7.2	8.2	7.3	8.3	7.4	7.8	6.8	—

Conductivity (umhos / cm)

Day														
Meter #	1	--	1	--	1	--	1	--	1	--	1	--	1	--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	319	--	318	--	316	--	316	--	315	--	319	--	320	--
13%	438	--	432	--	449	--	449	--	448	--	416	--	412	--
25%	541	--	549	--	569	--	571	--	566	--	498	--	497	--
50%	756	--	765	--	815	--	817	--	808	--	678	--	679	--
76%	992	--	997	--	1067	--	1070	--	1062	--	866	--	860	--
100%	1202	--	1200	--	1291	--	1297	--	1290	--	1033	--	1035	--
		--		--		--		--		--		--		--

Temperature (° C)

Day														
Meter #	3	3	3	0	3	5	3	3	3	5	5	5	5	
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	24	24	24	24	24	25	24	24	24	24	24	24	—
13%	24	24	24	24	24	24	25	24	24	24	24	24	24	—
25%	24	24	24	24	24	24	25	24	24	24	24	24	24	—
50%	24	24	24	24	24	24	25	24	24	24	24	24	24	—
76%	24	24	24	24	24	24	25	24	24	24	24	24	24	—
100%	24	24	24	24	24	24	25	24	24	24	24	24	24	—

I = Initial Chemistry F = Final Chemistry

Note: D.O. meter also used for temperature measurement unless otherwise noted.

Ceriodaphnia dubia - Chronic Toxicity Test
Survival and Reproduction Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 020217-2

Treatment (% Effluent)	Day No.	Replicate 1	2	3	4	5	6	7	8	9	10	Average Young/ Female	Number of Live Adults (% Sur.)	Average Young/ Female % CV
Control	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	5	3	4	5	4	5	6	6	5	5		10	
	4	7	5	5	9	7	9	8	8	7	7		10	
	5	—	—	—	—	—	—	—	—	—	—		10	
	6	14	14	11	15	16	15	13	17	16	15		10	
	7													
Totals:		26	22	20	29	27	29	27	31	28	27	26.6	(100)	12.4
# Broods (% 3rd Brood)		3	3	3	3	3	3	3	3	3	3	(100)		
13%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	6	6	6	5	3	3(1)	5	5(1)	7	6		10	
	4	7	6	7	7	6	10	12	6	10	10		10	
	5	—	—	—	—	—	—	—	—	—	—		10	
	6	14	15	16	17	15	8	15	13	15	15		10	
	7													
Totals:		27	27	29	29	24	21	32	24	32	31	27.6	(100)	13.5
25%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	8	4	6	5	5	6	5	7	6	5		10	
	4	—	8	12	14	10	8	12	16	~~120~~ 12	11		10	
	5	14	—	16	—	—	—	—	—	—	—		10	
	6	16	14	—	15	15	17	16	15	14	15		10	
	7													
Totals:		38	26	34	34	30	31	33	38	32	31	32.7	(100)	11.1
50%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	7	6	6	6	6	6	6	6	6	6		10	
	4	11	10	8	6	9	9	10	9	10	6		10	
	5	—	—	—	—	19	—	—	—	—	—		10	
	6	15	14	13	14	—	18	14	17	15	16		10	
	7													
Totals:		33	30	27	26	34	33	30	32	31	28	30.4	(100)	11.1
76%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	6	5	6	5	5	6	5	5	4	6		10	
	4	11	7	10	10	11	10	7	9	5	11		10	
	5	—	—	13	—	—	—	—	—	—	—		10	
	6	14	12	—	14	14	14	9	11	13	14		10	
	7													
Totals:		31	24	29	29	30	30	21	25	22	31	27.2	(100)	14.1
100%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	6	3	6	5	6	(E)	5	5	5	5		10	
	4	10	5	7	6	10	11	12	11	11	11		10	
	5	—	—	—	—	—	—	—	—	17	—		10	
	6	12	9	15	8	13	15	14	14	—	13		10	
	7													
Totals:		28	17	28	19	29	26	31	30	33	29	27.0	(100)	19.0

X = DEAD ADULT 1X = DEAD ADULT, ONE YOUNG PRODUCED BEFORE DEATH -- = NO YOUNG RECORDED
(E) = ABORTED EMBRYOS /EGGS (1) = ONE DEAD YOUNG (S) = SPLIT BROOD * = 4th BROOD EXCLUDED FROM TOTAL

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 020217-2

Brood Board Information:

Replicate	1	2	3	4	5	6	7	8	9	10	Brood Board Date: 01/25/17
Chamber Number	41	31	12	42	33	43	38	48	40	50	Young Age Range: 19-21 hours

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
YCT Batch #:	16-16	16-16	16-16	16-16	16-16	01-17	—	
Algae Batch #:	02-17	02-17	02-17	02-17	02-17	03-17	—	
Observation Time:	1100	0900	1500	1330	1330	1230	1130	
Initials:	CM	CM	AM	SR	CM	[illegible]	CM	
Date:	02/02/17	02/03/17	02/04/17	02/05/17	02/06/17	02/07/17	02/08/17	

Comment Section:

Day	Date	Initials	Comments

CETIS Analytical Report

Report Date: 10 Feb-17 13:59 (p 1 of 2)
Test Code: 61AEDA52 | 16-3884-9106

Ceriodaphnia 7-d Survival and Reproduction Test ERM

Analysis ID:	21-1052-2798	**Endpoint:**	Reproduction	**CETIS Version:**	CETISv1.8.7
Analyzed:	10 Feb-17 13:58	**Analysis:**	Parametric-Control vs Treatments	**Official Results:**	Yes

Batch ID:	16-0816-1722	**Test Type:**	Reproduction-Survival (7d)	**Analyst:**	Lab Tech
Start Date:	02 Feb-17 11:00	**Protocol:**	EPA/821/R-02-013 (2002)	**Diluent:**	Reconstituted Water
Ending Date:	08 Feb-17 11:30	**Species:**	Ceriodaphnia dubia	**Brine:**	
Duration:	6d 0h	**Source:**	In-House Culture	**Age:**	<24h

Sample ID:	16-4189-3280	**Code:**	61DD4DA0	**Client:**	City of Cadillac WWTP
Sample Date:	01 Feb-17 07:00	**Material:**	Industrial Effluent	**Project:**	WET Compliance Testing
Receive Date:	02 Feb-17 10:00	**Source:**	City of Cadillac WWTP		
Sample Age:	28h (1 °C)	**Station:**	Outfall 001A		

Data Transform	Zeta	Alt Hyp	Trials	Seed	PMSD	NOEL	LOEL	TOEL	TU
Untransformed	NA	C > T	NA	NA	14.6%	100	>100	NA	1.0 per WDGQ KM

Dunnett Multiple Comparison Test

Control	vs	C-%	Test Stat	Critical	MSD	DF	P-Value	P-Type	Decision(α:5%)
Lab Water		13	-0.5903	2.29	3.878	18	0.9525	CDF	Non-Significant Effect
		25	-3.601	2.29	3.878	18	1.0000	CDF	Non-Significant Effect
		50	-2.243	2.29	3.878	18	0.9998	CDF	Non-Significant Effect
		76	-0.3542	2.29	3.878	18	0.9176	CDF	Non-Significant Effect
		100	-0.2361	2.29	3.878	18	0.8941	CDF	Non-Significant Effect

Test Acceptability Criteria

Attribute	Test Stat	TAC Limits	Overlap	Decision
Control Resp	26.6	15 - NL	Yes	Passes Acceptability Criteria
PMSD	0.1458	0.13 - 0.47	Yes	Passes Acceptability Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	295.6833	59.13667	5	4.121	0.0030	Significant Effect
Error	774.9	14.35	54			
Total	1070.583		59			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Bartlett Equality of Variance	3.817	15.1	0.5760	Equal Variances
Distribution	Shapiro-Wilk W Normality	0.9531	0.946	0.0219	Normal Distribution

Reproduction Summary

C-%	Control Type	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	Lab Water	10	26.6	24.23	28.97	27	20	31	1.046	12.4%	0.0%
13		10	27.6	24.94	30.26	28	21	32	1.176	13.5%	-3.76%
25		10	32.7	30.11	35.29	32.5	26	38	1.146	11.1%	-22.9%
50		10	30.4	28.46	32.34	30.5	26	34	0.8589	8.93%	-14.3%
76		10	27.2	24.46	29.94	29	21	31	1.209	14.1%	-2.26%
100		10	27	23.34	30.66	28.5	17	33	1.619	19.0%	-1.5%

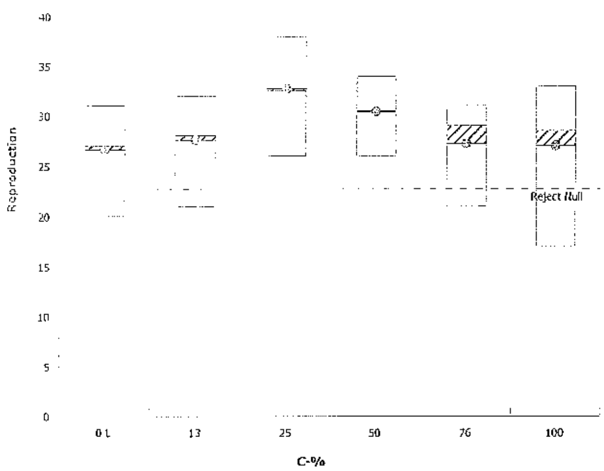
Reproduction Detail

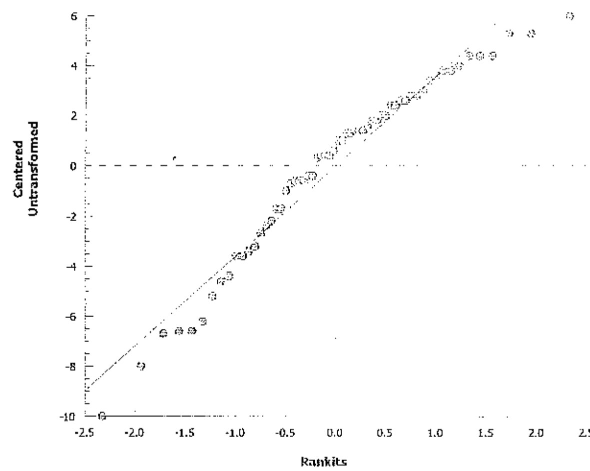
C-%	Control Type	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	Lab Water	26	22	20	29	27	29	27	31	28	27
13		27	27	29	29	24	21	32	24	32	31
25		38	26	34	34	30	31	33	38	32	31
50		33	30	27	26	34	33	30	32	31	28
76		31	24	29	29	30	30	21	25	22	31
100		28	17	28	19	29	26	31	30	33	29

Ceriodaphnia 7-d Survival and Reproduction Test — ERM

Analysis ID:	21-1052-2798	**Endpoint:**	Reproduction	**CETIS Version:**	CETISv1.8.7
Analyzed:	10 Feb-17 13:58	**Analysis:**	Parametric-Control vs Treatments	**Official Results:**	Yes

Graphics





ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	Taysha Laczko
ADDRESS:	1121 Plett Rd Cadillac, MI 49601	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert.	1-31-17 2-1-17	7AM 7AM	COMP	2 / GAL.	pH= 7.23 s.u. NH_3= 1.96 mg/L	020217-2	Temp. ✓ (°C) ☒ On Ice	D.O. 9.3 mg/L	pH 6.8 s.u.	Cond 1180 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
- ✓ Water/Wastewater
- ___ Sediment
- ___ Product

Test Type:
- ___ Acute
- ✓ Chronic
- ___ Other
- ___

Test Species:
- ✓ *Ceriodaphnia dubia*
- ___ *Daphnia magna*
- ___ *Daphnia pulex*
- ✓ Fathead minnow
- ___ *Americamysis bahia*
- ___ Sheepshead minnow
- ___ Silverside minnow
- ___ Rainbow Trout
- ___ *Chironomus dilutus*
- ___ *Hyalella azteca*
- ___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
Taysha Laczko / City of Cadillac	2-1-17	1:30PM	Colleen Nogel / ERM	02/02/17	1000

* See Instructions for Sample Collection on Back of Sheet

August 2010

ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	Taysha Laczko
ADDRESS:	1121 Plett Rd Cadillac, MI	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert.	2-2-17 2-3-17	7AM 6:55AM	comp	3 GAL	pH= 7.16 s.u. NH_3= .593 mg/L	020417-1	Temp. 1 (°C) ☐ On Ice	D.O. 9.9 mg/L	pH 7.1 s.u.	Cond 1364 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
- ✓ Water/Wastewater
- ___ Sediment
- ___ Product

Test Type:
- ___ Acute
- ✓ Chronic
- ___ Other

Test Species:
- ✓ *Ceriodaphnia dubia*
- ___ *Daphnia magna*
- ___ *Daphnia pulex*
- ✓ Fathead minnow
- ___ *Americamysis bahia*
- ___ Sheepshead minnow
- ___ Silverside minnow
- ___ Rainbow Trout
- ___ *Chironomus dilutus*
- ___ *Hyalella azteca*
- ___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
T. Laczko	2-3-17	12:50pm	[signature] /ERM	02/04/17	1030

* See Instructions for Sample Collection on Back of Sheet

August 2010

ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	Taysha Laczko
ADDRESS:	1121 Plett Rd Cadillac	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert	2-5-17 2-6-17	7AM 7AM	comp	2GAL	pH= 6.93 s.u. NH_3= .0476 mg/L	020717-5	Temp. [illegible] (°C) ☐ On Ice	D.O. 11.2 mg/L	pH 6.5 s.u.	Cond 1038 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
✓ Water/Wastewater
___ Sediment
___ Product

Test Type:
___ Acute
✓ Chronic
___ Other

Test Species:
✓ *Ceriodaphnia dubia*
___ *Daphnia magna*
___ *Daphnia pulex*
✓ Fathead minnow
___ *Americamysis bahia*
___ Sheepshead minnow
___ Silverside minnow
___ Rainbow Trout
___ *Chironomus dilutus*
___ *Hyalella azteca*
___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
T. Laczko	2-6-17	12:40pm	Colleen Nagel / ERM	02/07/17	1030

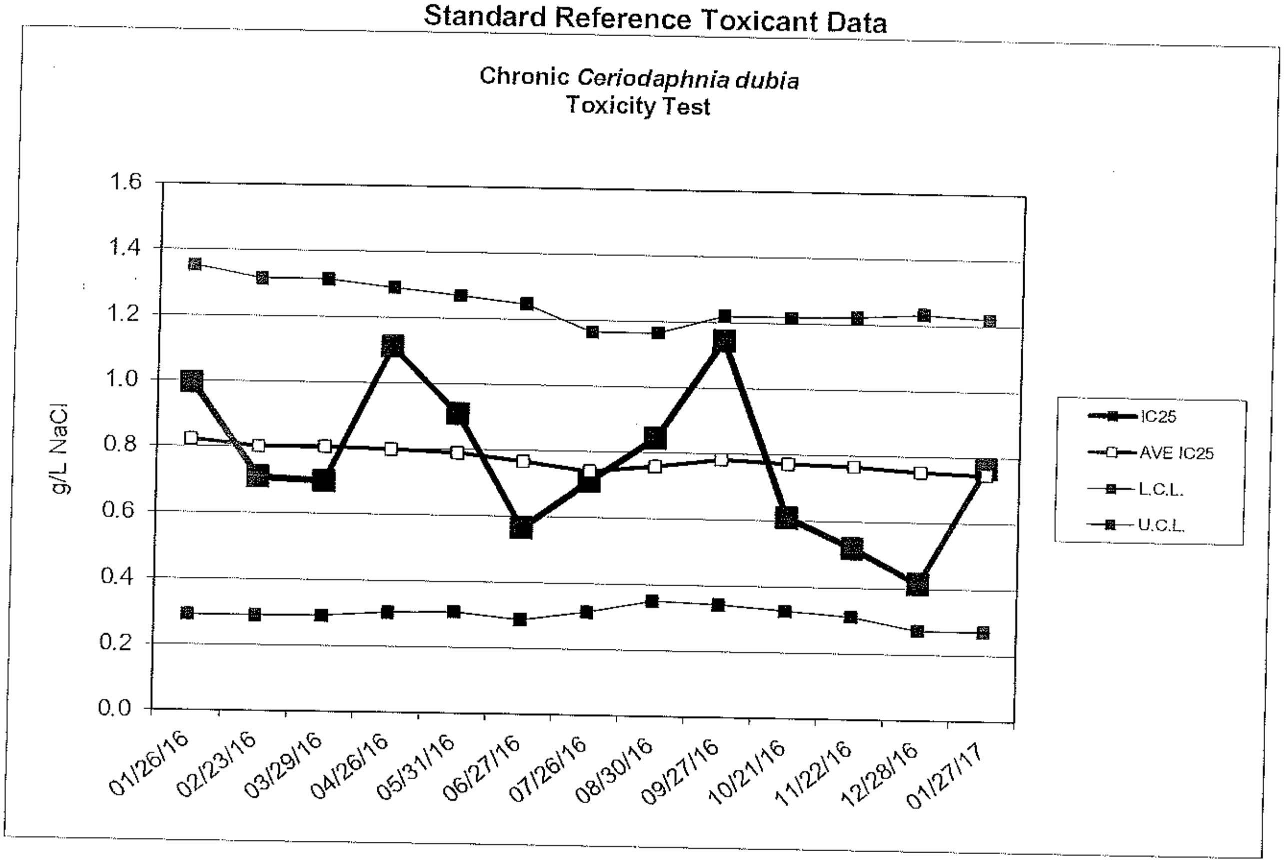
* See Instructions for Sample Collection on Back of Sheet

August 2010

Environmental Resources Management

ERM

Standard Reference Toxicant Data



Chronic *Ceriodaphnia dubia* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT Lower	Upper	Survival (%)	CONTROL Reproduction (ave. young)	CV (%)
01/26/16	1.0	0.8	0.3	1.4	100	28.4	9.7
02/23/16	0.7	0.8	0.3	1.3	100	26.5	20.4
03/29/16	0.7	0.8	0.3	1.3	100	33.2	6.3
04/26/16	1.1	0.8	0.3	1.3	100	28.6	23.6
05/31/16	0.9	0.8	0.3	1.3	100	34.1	9.2
06/27/16	0.6	0.8	0.3	1.2	100	31.0	20.2
07/26/16	0.7	0.7	0.3	1.2	100	31.2	11.1
08/30/16	0.8	0.8	0.4	1.2	100	27.7	9.2
09/27/16	1.1	0.8	0.3	1.2	100	24.1	24.2
10/21/16	0.6	0.8	0.3	1.2	100	26.9	15.1
11/22/16	0.5	0.8	0.3	1.2	100	28.0	17.4
12/28/16	0.4	0.8	0.3	1.2	100	20.5	22.6
01/27/17	0.8	0.7	0.3	1.2	100	24.9	22.6

Appendix B
MDEQ Reporting Form

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY – WATER BUREAU

CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than 1 test.

1. NAME OF FACILITY (on NPDES permit): City of Cadillac WWTP

2. NPDES PERMIT #:

M	I	0	0	2	0	2	5	7

1. RECEIVING WATER (as designated in permit): Clam River

2. OUTFALL: 001A

5. RECEIVING WATER CONCENTRATION (if known):

6. TEST LAB (Name and Address): Environmental Resources Management, 3352 128th Avenue, Holland, MI 49424-9263

7. TEST START DATE: 02/02/17

8. TEST END DATE: 02/08/17

9. AGE RANGE OF ORGANISMS AT TEST START: 19 - 21 Hours Old

10. REPORT DATE: 02/13/17

11. NAME OF PERSON CONDUCTING TEST: Lab Contact: Mark Baker

12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT: Bruce Rabe (616) 738 – 7308

13. SAMPLE COLLECTION DATES	14. DATE RECEIVED	15. ARRIVAL TEMP (C°)
Sample 1: 02/01/17	Sample 1: 02/02/17	Sample 1: 1
Sample 2: 02/03/17	Sample 2: 02/04/17	Sample 2: 1
Sample 3: 02/06/17	Sample 3: 02/07/17	Sample 3: 1

16. DATE OF FIRST USE	17. TOTAL RESIDUAL CHLORINE (mg/l)	18. AMMONIA (mg/l as N)
Sample 1: 02/02/17	Sample 1: <0.01	Sample 1: 2.5
Sample 2: 02/04/17	Sample 2: <0.01	Sample 2: 1.3
Sample 3: 02/07/17	Sample 3: <0.01	Sample 3: <0.25

19. WAS SAMPLE DECHLORINATED?

- Sample 1: ☐ YES ☒ NO IF YES ➡
- Sample 2: ☐ YES ☒ NO IF YES ➡
- Sample 3: ☐ YES ☒ NO IF YES ➡

20. DESCRIBE DECHLORINATION (if any)

21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE

22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time.)

23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡

24. STATE MESH SIZE OF FILTER

25. EFFLUENT SAMPLE TYPE (check one type for each sample)

- Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE
- Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE
- Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE

26. IDENTIFY THE DILUENT (O_1) CONTROL ____RMHW__

IDENTIFY THE SECONDARY (O_2) CONTROL (if used) ________________

27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTION

CONCENTRATION OF EFFLUENT (%)	O_1	O_2	11%	21%	42%	76%	100%
48-HOUR SURVIVAL (%)	100	N/A	100	100	100	100	100
6-DAY MEAN REPRODUCTION/FEMALE	26.6	N/A	27.6	32.7	30.4	27.2	27.0
6-DAY MEAN SURVIVAL (%)	100	N/A	100	100	100	100	100

28. 48-HOUR LC_{50} (%): >100

29. TU_a (acute toxic units): 0

30. 6-DAY CHRONIC VALUE (%): >100

31. NOEC: 100%

32. LOEC: >100%

33. TU_c (chronic toxic units): 0

EQP 5819 (02/10)

FINAL REPORT

Chronic Toxicity Test
Freshwater Vertebrate,
Pimephales promelas
EPA Test Method 1000.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Sample: Outfall 001A

Testing Period: 2 – 9 February 2017

Laboratory I.D. Number: 020217-2

PA DEP ID No. 68-04227
NJ DEP ID No. MI013

Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424

Test Overview

Permittee: City of Cadillac WWTP
Location: 1121 Plett Rd.
Cadillac, MI 49601
Contact: Mr. Derek Goodrich
Telephone #: 231.878.4437

NPDES Permit #: MI0020257
Permit Requirements: Acute Toxicity Limit = Report Only
Chronic Toxicity Limit = Report Only
Test Sample: Outfall 001A
Receiving Water: Clam River

Testing Date: 2 - 9 February 2017

Sample Date(s): 1 February 2017
3 February 2017
6 February 2017

Test/Method: Fathead Minnow, *Pimephales promelas*, Survival and Growth Test EPA 821-R-02-013 Method 1000.0.

QC Objectives: Test data met all test acceptability criteria, except where noted below.

Data Qualifiers: None

DATA SUMMARY

Effluent Concentrations (%)	Survival (%)	Growth Average Wt./ Organism (mg)
Control	87.5	0.470
13	100	0.502
25	87.5	0.485
50	100	0.506
76	97.5	0.509
100	100	0.521

TEST RESULTS

96-Hour LC_{50}	>100%
NOEC	100%
LOEC	>100%
ChV	>100%
MSDp	26.4%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	0

TEST CONCLUSION

In accordance with the NPDES permit requirements for City of Cadillac WWTP, this toxicity test did not exhibit acute or chronic toxicity.

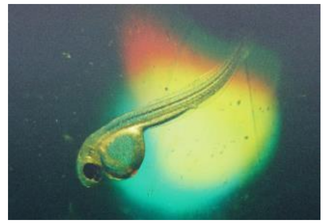
Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0343408.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777

ERM

ERM Testing Method

Pimephales promelas – Survival and Growth Toxicity Test



Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (µm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius (°C) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Pimephales promelas used to initiate this test were obtained from in-house cultures and were less than 24-hours old at test initiation. Test organisms were maintained in reconstituted moderately hard water prior to test initiation.

The *Pimephales promelas* test was conducted using 300 to 500-milliliter (mL) disposable polypropylene containers containing 250 mL of control water or test solution. Ten fish were randomly added to each test chamber with four replicate chambers per treatment. Each *Pimephales promelas* test chamber was fed 0.1 mL of a concentrated suspension of less than 24-hour old live brine shrimp nauplii (*Artemia* sp.) two times per day. Test solutions were renewed daily during the exposure by replacing approximately 90 percent of the 24-hour old solution with fresh control water or appropriate test solution. Prior to test solution renewal, uneaten and dead brine shrimp, along with other debris, were removed from the bottom of the test chambers.

Percent survival of exposed *Pimephales promelas* was determined daily by enumeration of live organisms. Mortality was defined as no body movement after gentle prodding. At the termination of the chronic test, larvae in each test chamber were counted, dried, and weighed to the nearest 0.01 milligram (mg) on an analytical balance.

The test was conducted at a temperature of 25 ± 1°C under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Pimephales promelas* survival and growth. An NOEC is defined as the highest effluent concentration which does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration which does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration.

Prior to the determination of any significant differences in *Pimephales promelas* survival and growth, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed, followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or growth) and converted to a chronic toxic unit (TUc) for reporting purposes by 100/ChV, unless the LOEC was greater than 100% in which case the TUc was reported as 0.

To evaluate acute toxicity, a 96-hour LC_{50} and corresponding 95 percent confidence interval were also calculated, where possible. The LC_{50} value estimate was determined by using one of the following statistical methods: graphical, Spearman-Karber, Trimmed Spearman-Karber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 96-hour LC_{50} value was converted to an acute toxic unit (TUa) by $100/LC_{50}$.

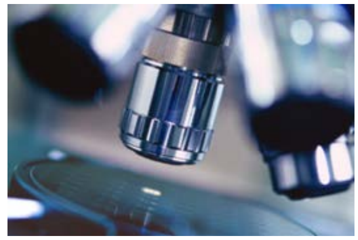
For 96-hour LC_{50} values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in the 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.8.7.15 software program

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC_{25}) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC_{25} of a particular reference toxicant test does not fall within the expected range of ± two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM'S STANDARD TEST METHOD

Test was performed in accordance with ERM's standard test method (see page 3).

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

Environmental Resources Management

Pimephales promelas **- Chronic Toxicity Test**
Initial Water Quality and Test Solution Preparation

Table 1
Page 1 of 1

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 020217-2
Beginning Date: 02/02/17 Time: 1130
Ending Date: 02/09/17 Time: 1130

Control/Dilution Water: RMHW
Organism Batch #: 21-17
Organism Age: <24 hrs
QC Review: KM
QC Review Date: 02/10/17

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	020217-2	020417-1	020717-5	06-17	—	—
Temperature	° C	1	1	1	--	--	--
pH	S.U.	6.8	7.1	6.5	7.8	—	—
Conductivity	umhos/cm	1180	1364	1038	317	—	—
Alkalinity	mg / L $CaCO_3$	120	140	100	58	—	—
Hardness	mg / L $CaCO_3$	220	240	460	86	—	—
Total Ammonia	mg / L NH_3	2.5	1.3	<0.25	--	--	--
Total Residual Chlorine	mg / L Cl_2	<0.01	<0.01	<0.01	--	--	--
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	CM	SPR	KM	CM	—	—

Test Solution Preparation: Test Solution Prepared For Both Species.

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)
Control	0	1200
13%	156	1044
25%	300	900
50%	600	600
76%	912	288
100%	1200	0

Test Day	Initials	Effluent Sample #	Synthetic Batch #
0	CM	1	06-17
1	CM	1	06-17
2	SPR	2	06-17
3	SPR	2	06-17
4	CM	~~3~~ 2	06-17 cm 02/06
5	RM	3	06-17
6	CM	3	06-17
7	RM	--	--

Environmental Resources Management

Pimephales promelas - Chronic Toxicity Test
Water Quality Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 020217-2

Dissolved Oxygen (mg/L) — Day

Meter #	3	3	3	3	3	3	3	3	3	5	5	3	5	5
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.5	7.3	8.5	6.4	8.2	6.5	8.1	6.5	8.5	6.9	8.1	7.1	8.3	7.3
13%	8.5	7.5	8.5	6.0	8.3	6.7	8.2	6.6	8.5	6.4	8.1	7.2	8.3	6.9
25%	8.5	7.3	8.5	6.2	8.3	6.7	8.3	7.0	8.6	6.2	8.2	7.1	8.3	6.6
50%	8.5	6.9	8.5	5.8	8.4	6.3	8.3	6.2	8.6	5.8	8.3	6.8	8.3	6.5
76%	8.5	7.1	8.5	5.7	8.5	6.0	8.3	5.8	8.6	5.8	8.4	6.7	8.2	5.9
100%	8.5	7.0	8.5	5.8	8.6	5.7	8.4	5.8	8.5	5.3	8.4	6.5	8.2	6.1

pH (S.U.) — Day

Meter #	9	8	8	8	4	4	4	8	8	9	9	8	9	9
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.8	7.5	7.8	7.5	7.6	7.5	7.4	7.5	7.8	7.3	7.9	7.6	7.8	7.5
13%	--	7.6	--	7.5	--	7.5	--	7.6	--	7.5	--	7.6	--	7.5
25%	--	7.7	--	7.6	--	7.6	--	7.6	--	7.5	--	7.6	--	7.4
50%	--	7.8	--	7.6	--	7.6	--	7.6	--	7.5	--	7.5	--	7.4
76%	--	7.8	--	7.7	--	7.7	--	7.6	--	7.5	--	7.3	--	7.4
100%	7.2	8.0	7.2	7.8	7.3	7.8	7.2	7.7	7.3	7.6	7.4	7.4	6.8	7.3

Conductivity (umhos / cm) — Day

Meter #	1	--	1	--	1	--	1	--	1	--	1	--	1	--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	319	--	318	--	316	--	318	--	315	--	319	--	320	--
13%	438	--	432	--	449	--	449	--	448	--	410	--	412	--
25%	541	--	549	--	569	--	571	--	566	--	498	--	497	--
50%	756	--	765	--	815	--	817	--	808	--	678	--	679	--
76%	992	--	997	--	1067	--	1070	--	1062	--	866	--	860	--
100%	1202	--	1200	--	1291	--	1297	--	1290	--	1033	--	1035	--

Temperature (° C) — Day

Meter #	3	3	3	8J DM	3	3	3	3	3	5	5	3	5	5
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	24	24	25	24	25	25	25	24	25	24	24	24	24
13%	24	25	24	25	24	25	25	23	24	25	24	24	24	25
25%	24	25	24	25	24	25	25		24	25	24	25	24	25
50%	24	25	24	25	24	25	25		24	24	24	24	24	25
76%	24	24	24	25	24	26	25	25	24	25	24	25	24	25
100%	24	24	24	25	24	26	25	25	24	25	24	25	24	25

I = Initial Chemistry F = Final Chemistry

Note: D.O. meter also used for temperature measurement unless otherwise noted.

Environmental Resources Management

Pimephales promelas - Chronic Toxicity Test
Survival Data

Table 3
Page 1 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 020217-2

Treatment (% Effluent)	Rep.	# Live Organisms Day 0	1	2	3	4	5	6	7	Rep.	# Live Organisms Day 0	1	2	3	4	5	6	7	96 Hour Survival Summary Total Live Initial	Final	% Survival
Control	A	10	10	10	9	9	8	7	7	B	10	10	10	10	10	10	10	10	40	38	95
13%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100
25%	A	10	9	9	9	9	9	9	9	B	10	10	10	10	10	10	10	9	40	36	90
50%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100
76%	A	10	10	10	9	9	9	9	9	B	10	10	10	10	10	10	10	10	40	39	97.5
100%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100

Treatment (% Effluent)	Rep.	# Live Organisms Day 0	1	2	3	4	5	6	7	Rep.	# Live Organisms Day 0	1	2	3	4	5	6	7	7 Day Survival Summary Total Live Initial	Final	% Survival
Control	C	10	10	10	10	10	10	10	10	D	10	10	10	9	9	9	8	8	40	35	87.5
13%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100
25%	C	10	10	9	9	9	9	9	9	D	10	10	10	9	8	8	8	8	40	35	87.5
50%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100
76%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	39	97.5
100%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Time:	1130	0930	1500	1530	1400	1200	1130	1130
Initials:	CM	CM	[illegible]	SPK	CM	RL	CM	RL
Date:	02/02/17	02/03/17	02/04/17	02/05/17	02/06/17	02/07/17	02/08/17	02/09/17

Comment Section:

Day	Date	Initials	Comments

Environmental
Resources
Management

Pimephales promelas - Chronic Toxicity Test
Growth Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 020217--2

Pan #	Conc. (% Effluent)	Replicate	Final Weight (mg)	Initial Weight (mg)	Larvae Weight (mg)	# of Initial Organisms	Avg. Wt./ Organism/ Replicate (mg)	Avg. Wt./ Organism/ Treatment (mg)	Avg. Wt./ Organism/ Treatment % CV
Date			02/10/17	02/08/17					
Analyst			cn	mb					
1	Control	A	30.25	26.57	3.68	10	0.368		
2	Control	B	31.26	25.94	5.32	10	0.532		
3	Control	C	22.76	16.74	6.02	10	0.602		
4	Control	D	24.76	20.99	3.77	10	0.377	0.470	24.7
5	13%	A	28.14	22.62	5.52	10	0.552		
6	13%	B	21.87	16.76	5.11	10	0.511		
7	13%	C	21.83	16.71	5.12	10	0.512		
8	13%	D	19.07	14.74	4.33	10	0.433	0.502	9.9
9	25%	A	25.45	20.47	4.98	10	0.498		
10	25%	B	23.52	18.67	4.85	10	0.485		
11	25%	C	26.88	21.61	5.27	10	0.527		
12	25%	D	24.83	20.53	4.30	10	0.430	0.485	8.4
13	50%	A	24.95	19.51	5.44	10	0.544		
14	50%	B	27.77	23.12	4.65	10	0.465		
15	50%	C	23.55	17.65	5.90	10	0.590		
16	50%	D	21.46	17.23	4.23	10	0.423	0.506	14.9
17	76%	A	22.67	18.75	3.92	10	0.392		
18	76%	B	25.28	19.93	5.35	10	0.535		
19	76%	C	26.06	20.20	5.86	10	0.586		
20	76%	D	27.53	22.31	5.22	10	0.522	0.509	16.2
21	100%	A	22.15	16.35	5.80	10	0.580		
22	100%	B	21.66	16.76	4.90	10	0.490		
23	100%	C	24.41	19.21	5.20	10	0.520		
24	100%	D	28.49	23.54	4.95	10	0.495	0.521	7.9

Quality Assurance Final Wt. (mg)

25	Blank	A	9.65	9.64	0.01
26	Blank	B	11.65	11.65	0.00

CETIS Analytical Report

Fathead Minnow 7-d Larval Survival and Growth Test — ERM

Analysis ID:	13-2784-0912	Endpoint:	Mean Dry Biomass-mg	CETIS Version:	CETISv1.8.7
Analyzed:	10 Feb-17 13:11	Analysis:	Parametric-Control vs Treatments	Official Results:	Yes
Batch ID:	01-0064-2024	Test Type:	Growth-Survival (7d)	Analyst:	Lab Tech
Start Date:	02 Feb-17 11:30	Protocol:	EPA/821/R-02-013 (2002)	Diluent:	Reconstituted Water
Ending Date:	09 Feb-17 11:30	Species:	Pimephales promelas	Brine:	
Duration:	7d 0h	Source:	In-House Culture	Age:	<24h
Sample ID:	15-2579-1072	Code:	5AF1B960	Client:	City of Cadillac WWTP
Sample Date:	01 Feb-17 07:00	Material:	Industrial Effluent	Project:	WET Compliance Testing
Receive Date:	02 Feb-17 10:00	Source:	City of Cadillac WWTP		
Sample Age:	28h (1 °C)	Station:	Outfall 001A		

Data Transform	Zeta	Alt Hyp	Trials	Seed	PMSD	NOEL	LOEL	TOEL	TU
Untransformed	NA	C > T	NA	NA	26.4%	100	>100	NA	~~1~~ 0

per MDEQ KM

Dunnett Multiple Comparison Test

Control	vs	C-%	Test Stat	Critical	MSD	DF	P-Value	P-Type	Decision(α:5%)
Lab Water		13	-0.6266	2.41	0.124	6	0.9551	CDF	Non-Significant Effect
		25	-0.2963	2.41	0.124	6	0.9058	CDF	Non-Significant Effect
		50	-0.6946	2.41	0.124	6	0.9620	CDF	Non-Significant Effect
		76	-0.7577	2.41	0.124	6	0.9675	CDF	Non-Significant Effect
		100	-1.001	2.41	0.124	6	0.9828	CDF	Non-Significant Effect

Test Acceptability Criteria

Attribute	Test Stat	TAC Limits	Overlap	Decision
Control Resp	0.4698	0.25 - NL	Yes	Passes Acceptability Criteria
PMSD	0.2637	0.12 - 0.3	Yes	Passes Acceptability Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	0.006769699	0.00135394	5	0.2555	0.9314	Non-Significant Effect
Error	0.09537122	0.005298401	18			
Total	0.1021409		23			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Bartlett Equality of Variance	4.786	15.1	0.4425	Equal Variances
Distribution	Shapiro-Wilk W Normality	0.9785	0.884	0.8676	Normal Distribution

Mean Dry Biomass-mg Summary

C-%	Control Type	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	Lab Water	4	0.4698	0.2853	0.6542	0.4545	0.368	0.602	0.05797	24.7%	0.0%
13		4	0.502	0.4227	0.5813	0.5115	0.433	0.552	0.0249	9.92%	-6.87%
25		4	0.485	0.4203	0.5497	0.4915	0.43	0.527	0.02033	8.38%	-3.25%
50		4	0.5055	0.3855	0.6255	0.5045	0.423	0.59	0.03771	14.9%	-7.61%
76		4	0.5088	0.3773	0.6402	0.5285	0.392	0.586	0.04129	16.2%	-8.3%
100		4	0.5212	0.4555	0.587	0.5075	0.49	0.58	0.02065	7.92%	-11.0%

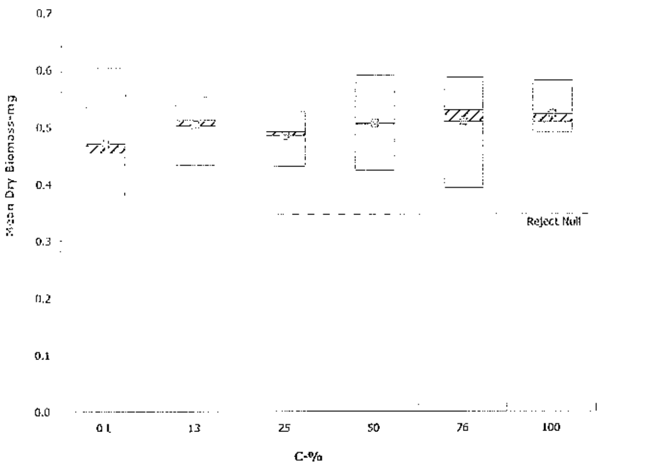
Mean Dry Biomass-mg Detail

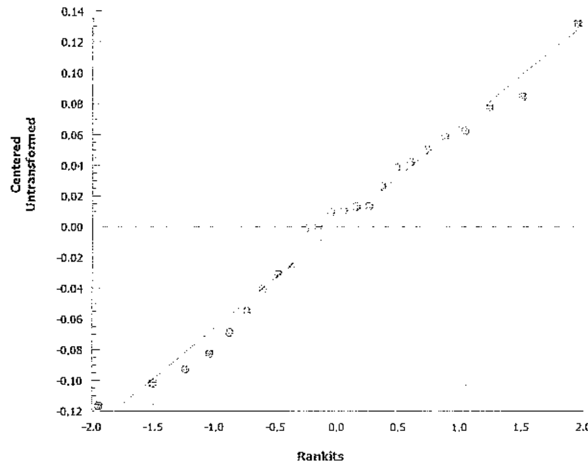
C-%	Control Type	Rep 1	Rep 2	Rep 3	Rep 4
0	Lab Water	0.368	0.532	0.602	0.377
13		0.552	0.511	0.512	0.433
25		0.498	0.485	0.527	0.43
50		0.544	0.465	0.59	0.423
76		0.392	0.535	0.586	0.522
100		0.58	0.49	0.52	0.495

Fathead Minnow 7-d Larval Survival and Growth Test ERM

Analysis ID:	13-2784-0912	**Endpoint:**	Mean Dry Biomass-mg	**CETIS Version:**	CETISv1.8.7
Analyzed:	10 Feb-17 13:11	**Analysis:**	Parametric-Control vs Treatments	**Official Results:**	Yes

Graphics





ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	Taysha Laczko
ADDRESS:	1121 Plett Rd Cadillac, MI 49601	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert.	1-31-17 2-1-17	7AM 7AM	comp	2 / GAL.	pH= 7.23 s.u. NH_3= 1.96 mg/L	02017-2	Temp. [illegible] (° C) ☒ On Ice	D.O. 9.3 mg/L	pH 6.8 s.u.	Cond 1180 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
- ✓ Water/Wastewater
- ___ Sediment
- ___ Product

Test Type:
- ___ Acute
- ✓ Chronic
- ___ Other

Test Species:
- ✓ *Ceriodaphnia dubia*
- ___ *Daphnia magna*
- ___ *Daphnia pulex*
- ✓ Fathead minnow
- ___ *Americamysis bahia*
- ___ Sheepshead minnow
- ___ Silverside minnow
- ___ Rainbow Trout
- ___ *Chironomus dilutus*
- ___ *Hyalella azteca*
- ___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
Taysha Laczko / City of Cadillac	2-1-17	1:30 PM	Colleen Nagel / ERM	02/02/17	1000

* See Instructions for Sample Collection on Back of Sheet

August 2010

ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	Taysha Laczko
ADDRESS:	1121 Plett Rd Cadillac, MI	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
							Temp. (°C)	D.O.	pH	Cond
Tert.	2-2-17 2-3-17	7AM 6:55AM	COMP	3 GAL	pH= 7.16 s.u. NH_3= .593 mg/L	020417-1	1 (°C) ☐ On Ice	9.9 mg/L	7.1 s.u.	1364 umhos/cm
					pH= s.u. NH_3= mg/L		(°C) ☐ On Ice	mg/L	s.u.	umhos/cm
					pH= s.u. NH_3= mg/L		(°C) ☐ On Ice	mg/L	s.u.	umhos/cm
					pH= s.u. NH_3= mg/L		(°C) ☐ On Ice	mg/L	s.u.	umhos/cm
					pH= s.u. NH_3= mg/L		(°C) ☐ On Ice	mg/L	s.u.	umhos/cm
					pH= s.u. NH_3= mg/L		(°C) ☐ On Ice	mg/L	s.u.	umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
- ✓ Water/Wastewater
- ___ Sediment
- ___ Product

Test Type:
- ___ Acute
- ✓ Chronic
- ___ Other
- ___

Test Species:
- ✓ Ceriodaphnia dubia
- ___ Daphnia magna
- ___ Daphnia pulex
- ✓ Fathead minnow
- ___ Americamysis bahia
- ___ Sheepshead minnow
- ___ Silverside minnow
- ___ Rainbow Trout
- ___ Chironomus dilutus
- ___ Hyalella azteca
- ___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
T. Laczko	2-3-17	12:50PM	[signature] / ERM	02/04/17	1030

* See Instructions for Sample Collection on Back of Sheet

August 2010

ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	Taysha Laczko
ADDRESS:	1121 Plett Rd Cadillac	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert	2-5-17 2-6-17	7AM 7AM	comp	2GAL	pH= 6.93 s.u. NH3= .0476 mg/L	020717-5	Temp. 20.1 (° C) ☐ On Ice	D.O. 11.2 mg/L	pH 6.5 s.u.	Cond 1038 umhos/cm
					pH= s.u. NH3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (° C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
- ✓ Water/Wastewater
- ___ Sediment
- ___ Product

Test Type:
- ___ Acute
- ✓ Chronic
- ___ Other

Test Species:
- ✓ *Ceriodaphnia dubia*
- ___ *Daphnia magna*
- ___ *Daphnia pulex*
- ✓ Fathead minnow
- ___ *Americamysis bahia*
- ___ Sheepshead minnow
- ___ Silverside minnow
- ___ Rainbow Trout
- ___ *Chironomus dilutus*
- ___ *Hyalella azteca*
- ___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
T. Laczko	2-6-17	12:40pm	Colleen Nagel / ERM	02/07/17	1030

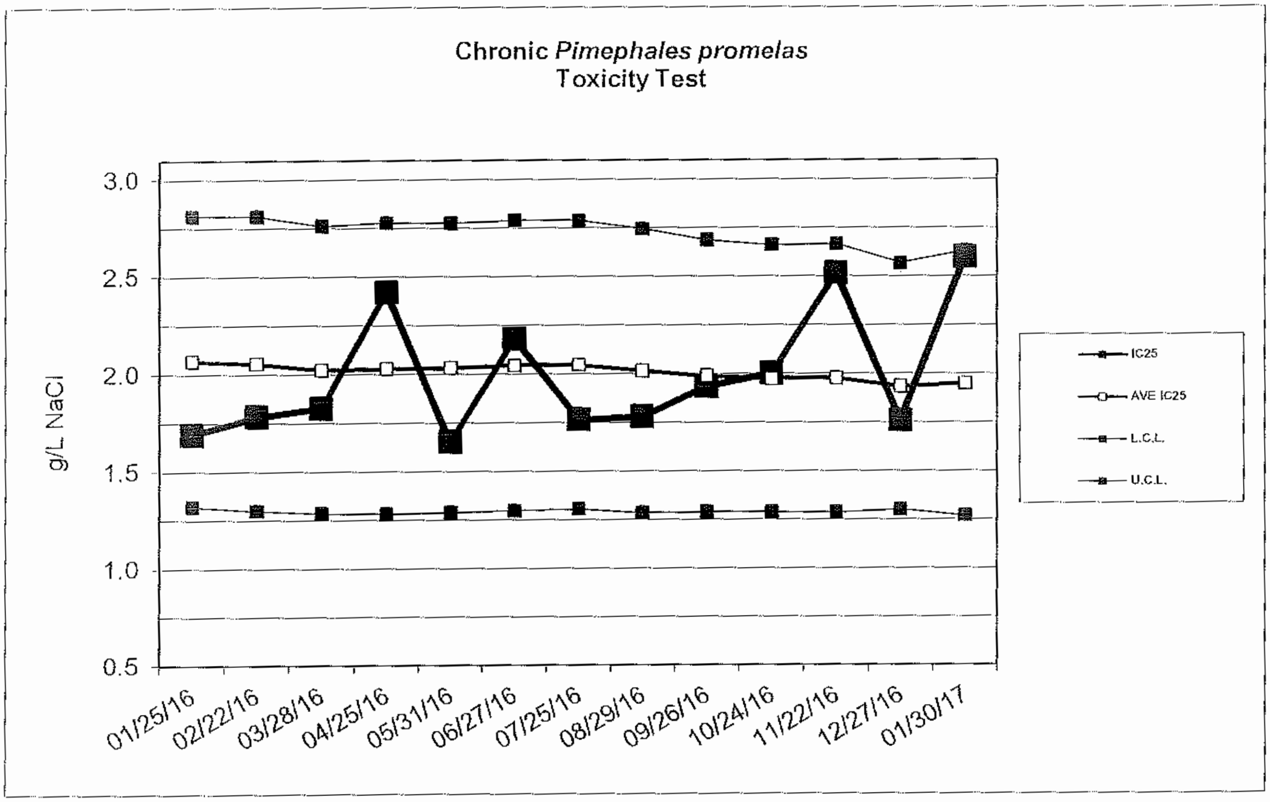
* See Instructions for Sample Collection on Back of Sheet

August 2010

Environmental Resources Management

ERM

Standard Reference Toxicant Data



Chronic *Pimephales promelas* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT Lower	CONTROL LIMIT Upper	Survival (%)	CONTROL Growth (mg)	CV (%)
01/25/16	1.7	2.1	1.3	2.8	97.5	0.55	10.3
02/22/16	1.8	2.1	1.3	2.8	95	0.63	6.0
03/28/16	1.8	2.0	1.3	2.8	95	0.36	23.2
04/25/16	2.4	2.0	1.3	2.8	80	0.53	17.3
05/31/16	1.7	2.0	1.3	2.8	97.5	0.53	12.3
06/27/16	2.2	2.0	1.3	2.8	100	0.54	9.0
07/25/16	1.8	2.0	1.3	2.8	100	0.45	6.0
08/29/16	1.8	2.0	1.3	2.7	100	0.57	1.7
09/26/16	1.9	2.0	1.3	2.7	97.5	0.63	8.1
10/24/16	2.0	2.0	1.3	2.7	100	0.60	18.6
11/22/16	2.5	2.0	1.3	2.7	90	0.64	9.6
12/27/16	1.8	1.9	1.3	2.6	95	0.43	6.7
01/30/17	2.6	2.0	1.3	2.6	80	0.51	32.2

Appendix B
MDEQ Reporting Form

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY – WATER BUREAU

FATHEAD MINNOW CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than one test.

1. NAME OF FACILITY (on NPDES permit): City of Cadillac WWTP

2. NPDES PERMIT #: M I 0 0 2 0 2 5 7

3. RECEIVING WATER (as designated in permit): Clam River

4. OUTFALL: 001A

5. RECEIVING WATER CONCENTRATION (if known):

6. TEST LAB (Name and Address): Environmental Resources Management, 3352 128th Avenue, Holland, MI 49424-9263

7. TEST START DATE: 02/02/17

8. TEST END DATE: 02/09/17

9. AGE RANGE OF ORGANISMS AT TEST START: <24 Hrs Old

10. REPORT DATE: 02/13/17

11. NAME OF PERSON CONDUCTING TEST: Lab Contact: Mark Baker

12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT: Bruce Rabe (616) 738 – 7308

13. SAMPLE COLLECTION DATES	14. DATE RECEIVED	15. ARRIVAL TEMP (C°)
Sample 1: 02/01/17	Sample 1: 02/02/17	Sample 1: 1
Sample 2: 02/03/17	Sample 2: 02/04/17	Sample 2: 1
Sample 3: 02/06/17	Sample 3: 02/07/17	Sample 3: 1

16. DATE OF FIRST USE	17. TOTAL RESIDUAL CHLORINE (mg/l)	18. AMMONIA (mg/l as N)
Sample 1: 02/02/17	Sample 1: <0.01	Sample 1: 2.5
Sample 2: 02/04/17	Sample 2: <0.01	Sample 2: 1.3
Sample 3: 02/07/17	Sample 3: <0.01	Sample 3: <0.25

19. WAS SAMPLE DECHLORINATED?

Sample 1: ☐ YES ☒ NO IF YES ➡

Sample 2: ☐ YES ☒ NO IF YES ➡

Sample 3: ☐ YES ☒ NO IF YES ➡

20. DESCRIBE DECHLORINATION (if any) N/A

21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE

22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time. Attach a separate sheet if necessary.) N/A

23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡

24. STATE MESH SIZE OF FILTER (if filtered)

25. EFFLUENT SAMPLE TYPE (check one type for each sample)

Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE

Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE

Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE

26. IDENTIFY THE DILUENT (O_1) CONTROL: RMHW

IDENTIFY THE SECONDARY (O_2) CONTROL (if used): ________

27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND GROWTH

CONCENTRATION OF EFFLUENT (%)	O_1 (dilution H_2O)	O_2	13%	25%	50%	76%	100%
96-HOUR SURVIVAL (%)	95	N/A	100	90	100	97.5	100
7-DAY MEAN BIOMASS (mg/initial fish)	0.470	N/A	0.502	0.485	0.506	0.509	0.521
7-DAY MEAN SURVIVAL (%)	87.5	N/A	100	87.5	100	97.5	100

28. 96-HOUR LC_{50} (%): >100

29. TU_a (acute toxic units): 0

30. 7-DAY CHRONIC VALUE (%): >100

31. NOEC: 100%

32. LOEC: >100%

33. TU_c (chronic toxic units): 0

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